

Project Title: OTT User Subscription Funnel Analysis

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Executive Summary

User engagement and subscription completion are critical for revenue growth in OTT platforms. However, a significant percentage of users drop off at different steps of the subscription journey. This project analyzes the funnel flow from Visit → Account Creation → Plan Selection → Payment Initiation → Payment Success to identify where users churn the most, understand the root causes, and propose data-driven business improvements to increase conversion.

Business Problem

Although a large number of users visit the OTT platform and view content or subscription plans, only a small percentage complete the payment and become paid subscribers. The major challenge is to understand at which stage users exit the funnel and why. Without this understanding, marketing spend is wasted, and revenue potential is not fully realized.

Objectives

- Identify funnel stages with the highest user drop-off.
 - Determine behavioral patterns influencing conversion.
 - Create performance metrics and visualizations for funnel monitoring.
 - Provide business recommendations to increase customer conversion.
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Dataset Overview

The dataset consists of OTT platform user interaction data including:

- User ID
- Gender, Age Group, Device Type
- Subscription Plan Viewed
- Login & Interaction Logs
- Subscription Attempt & Completion Status

Time Period: 6 Months of Platform Interaction Logs.

Tools and Technologies Used

- Python: Pandas, NumPy, Matplotlib
- Power BI: Data Modeling, DAX Measures, Funnel Chart, Slicers
- Excel: Initial Data Understanding

Methodology

1. Data Cleaning in Python

- Removed duplicates & invalid rows.
- Standardized date formats.
- Handled missing demographics using conditional imputation.
- Feature engineered steps such as "Viewed Plan", "Attempted Payment", "Payment Completed".

2. Funnel Construction in Python

- Grouped users based on funnel stage.
- Calculated % conversion between each stage.
- Visualized the funnel using bar & funnel charts.

3. Exporting Clean Data for Power BI

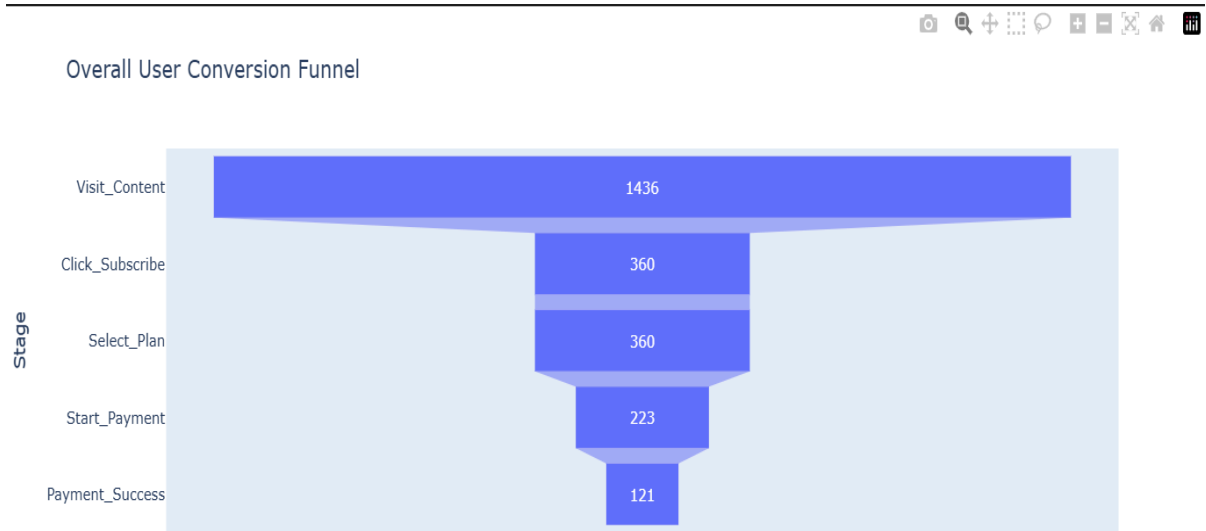
- Saved processed dataset as cleaned_ott_funnel_data.csv.

4. Power BI Dashboard

- Created relationships & data model.
- Created DAX measures:
 - Conversion %
 - Drop-Off %
 - Stage User Count
- Designed visual funnel & segment charts.
- Added slicers: Age, Gender, Device Type, Region & Subscription Plan.



Key Funnel Insights



Funnel Stage	Users	Drop %	Insight
Visited Platform	1436	–	High awareness; users browse content but many do not show direct subscription intent
Click Subscribe	360	74.9%	Major drop-off value proposition or CTA not compelling enough. Users may not see why they should subscribe.
Select Plan	360	0%	Users who click subscribe are willing to explore plans; plan information is understandable.
Start Payment	223	38.1%	Payment hesitation due to pricing sensitivity, trust concerns, or lack of preferred payment options.
Payment Completed	121	45.7%	High drop due to checkout friction, possible payment failures, or authentication issues.

Biggest drop-off occurs at the Payment Stage (Payment Initiation → Payment Success).

Business Insights

- A significant portion of users show interest but do not proceed to create an account.
 - Users who attempt signup but drop during payment indicate trust issues or payment gateway friction.
 - Younger users (18–24) show higher interaction but lower conversion, suggesting price sensitivity.
 - Mobile users convert less than TV/Desktop users, indicating UI/UX improvement potential.
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Recommendations

1. Improve Signup Flow
 - Reduce form fields, enable social login.
 2. Enhance Payment Experience
 - Offer UPI/autopay discounts.
 - Display secure payment badges.
 3. Targeted Remarketing
 - Send personalized reminders for abandoned users within 24 hours.
 4. Introduce 7-Day Trial for First-Time Users
 - Reduces hesitation and builds trust.
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Next Steps

- Perform A/B testing on signup and payment page UI (*A/B testing means showing two different versions of a screen to different user groups to see which version results in higher signup or payment completion*).
 - Track conversion per marketing campaign source.
 - Implement monthly funnel review dashboards.
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Conclusion

By analyzing user drop-off behavior across the funnel, we identified key friction points primarily at the signup and payment stages. Addressing these through optimized UX, trust-building payment communication, and targeted re-engagement strategies will significantly increase subscription conversion and revenue growth for the OTT platform.